



Strings

CS 101- Programming
Fundamentals

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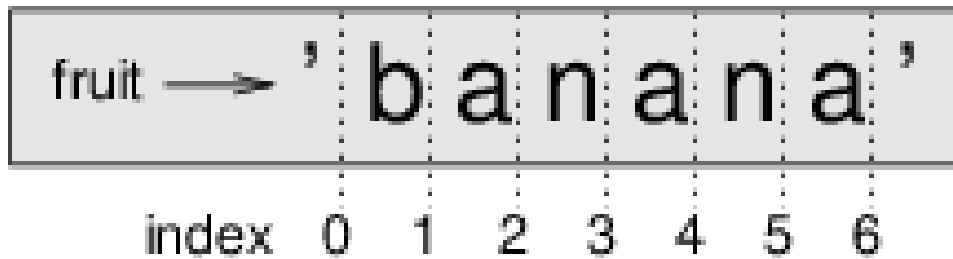
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String

- **Sequence of characters:** Not like integers, floats, and booleans.
- An ordered collection of other values.
- Immutable
- **Bracket operator:** used to access the characters one at a time.
- **Index:** indicates the position of character in the sequence.
 - [1], [0], [a], [a+1], [-1]
 - [1.5]

Slicing

- Slice: A segment of a string.
- **[n:m]**: returns the part of the string from the “n-eth” character (inclusive) to the “m-eth” character (exclusive).



- `>>> fruit = 'banana'`
- `>>> fruit[:3]`
- `'ban'`
- `>>> fruit[3:]`
- `'ana'`

- `>>> fruit = 'banana'`
- `>>> fruit[3:3]`
- `''`
-

String methods

- **Len:** built-in function that returns the number of characters in a string.
 - *IndexError: string index out of range*
- **Method invocation:** method name(dot) string name (parantheses)
 - **stringname.upper():** takes a string and returns a new string with all uppercase letters.
 - **stringname.lower():** takes a string and returns a new string with all lowercase letters.

String methods

- **find:** Finds characters, substrings within the string.
- **Syntax:** stringname(dot)find(character/substring to search)
- Optional arguments: index where it should start, index where it should stop

- `>>> 'Hello World'.find('llo')`
2

- `>>> 'Hello World'.find('o')`
4

- `>>> 'Hello World'.find('o', 4)`
7

- `>>> 'Hello World'.find('l', 4, 8)`
-1

String membership test: in operator

- **in:** Boolean operator that takes two strings and returns True if the first appears as a substring in the second.

```
word = 'Hello'  
word2 = 'World'  
for letter in word:  
    if letter in word2:  
        print (letter)
```

String comparison

- Using relational operators
- `==`, `!=`: Useful for checking equality or inequality
- `<`, `>`: useful for putting words in alphabetical order

To-do

- Read up other common string methods and their uses.
- How to use escape sequences to print special characters within a string?